

VARSHA PIRATI

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PROFESSIONAL SUMMARY

AI/ML Engineer and Data Science professional with around 1+ year of experience designing and implementing machine learning models, predictive analytics solutions, and data-driven applications. Proficient in Python, SQL, data analysis, statistical modeling, and machine learning techniques, with hands-on experience in recommendation systems, computer vision projects, and large-scale data processing. Skilled in data cleaning, exploratory data analysis (EDA), feature engineering, model evaluation, and translating business requirements into impactful AI solutions.

CORE COMPETENCIES:

- Machine Learning & Data Science
- Programming & Software Development
- Databases & Data Engineering
- Data Analysis & Visualization
- Tools, Platforms & Agile Methodologies

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, C++, C

AI/ML & Data Science: Machine Learning, Data Analysis, Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Model Evaluation, Statistical Analysis, Computer Vision, Recommendation Systems

Backend & Software Development: REST APIs, Spring Boot, API Testing (Postman), Object-Oriented Programming

Databases: MySQL, PostgreSQL, MongoDB, Database Design, Query Optimization

Cloud: AWS, Docker, Kubernetes, CI/CD Pipelines, GitHub Actions

Data Engineering: ETL Pipelines, Apache Airflow, Kafka, Data Processing, Data Analytics

Software Engineering: Data Structures & Algorithms, Debugging, Multithreading, Agile/Scrum, SDLC

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Jupyter Notebook

EXPERIENCE

Brane enterprises - Hyderabad, India

Jul 2023 - Jul 2024

AI/ML & Data Applications Developer

- Developed **Python-based backend services, REST APIs**, and data-driven applications to support analytics workflows and business reporting.
- Built **ETL pipelines** for ingesting, validating, cleaning, and transforming structured and semi-structured data using **Python, Pandas, and SQL**.
- Performed **Exploratory Data Analysis (EDA)**, **data preprocessing**, and **feature engineering** to improve data quality and support machine learning initiatives.
- Assisted in developing and evaluating **Machine Learning models** using **Scikit-learn** and **TensorFlow**, conducting model testing, validation, and performance analysis.
- Designed and optimized **PostgreSQL** and **MySQL** databases, improving **query performance** and supporting analytical workloads.
- Collaborated with cross-functional teams to integrate **AI/ML-driven features, recommendation systems**, and data analytics capabilities into business applications.
- Utilized **AWS** and **Docker** to support scalable development, testing, and deployment environments.
- Participated in **Agile/Scrum** processes including sprint planning, code reviews, testing, debugging, and release support activities.

Technologies: Python, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, REST APIs, MySQL, MongoDB, AWS, Docker, ETL Pipelines, Data Analytics, Machine Learning, Git, Agile/Scrum.

Software Developer & Automation Tester Intern - India

Oct 2022 - May 2023

- Developed and executed **automated test scripts using Selenium and Java**, improving software quality and reducing manual testing efforts.
- Worked with **backend systems, SQL databases, and large datasets** to validate application functionality, data integrity, and system performance.
- Performed **functional, regression, and API testing** to ensure application reliability across multiple releases.
- Investigated application defects, conducted **root-cause analysis**, and collaborated with development teams to resolve issues efficiently.
- Assisted in validating **REST APIs** using Postman and supported end-to-end testing of application workflows.
- Collaborated within **Agile/Scrum** teams, participating in sprint planning, testing, verification, and deployment support activities.
- Contributed to improving software stability and performance through debugging, issue tracking, and process optimization.

Technologies: Java, Selenium, SQL, REST APIs, Postman, MySQL, Git, Agile/Scrum, Functional Testing, Regression Testing, Debugging, Quality Assurance

ACADEMIC PROJECTS

Object Recognition System

- Developed an object detection system using YOLO, TensorFlow, and OpenCV to accurately identify and localize multiple objects in images.
- Performed image preprocessing, model training, and hyperparameter tuning to improve detection accuracy and optimize real-time inference performance.
- Evaluated model performance using computer vision techniques and integrated the trained model into a user-friendly interface for object recognition.

Technologies: Python, YOLO, TensorFlow, OpenCV, Machine Learning, Computer Vision, HTML

Multi-Resolution-System

- Developed an intelligent web application to analyze website responsiveness across multiple screen resolutions and device sizes using automated UI validation techniques.
- Built a data processing pipeline to collect, analyze, and visualize responsiveness metrics, enabling identification of layout inconsistencies and performance bottlenecks.
- Automated generation of analytical reports to support data-driven optimization, reducing manual testing effort and improving cross-device compatibility.

Technologies: Java, Python, SQL, HTML, CSS, Data Analysis, Machine Learning, Computer Vision, OpenCV

Image Caption Generation System

- Built an end-to-end machine learning pipeline for automatically generating descriptive captions from images.
- Performed data preprocessing, feature extraction, model training, and evaluation to improve prediction accuracy.
- Analysed model performance and optimized inference workflows for efficient caption generation.

Technologies: Python, Machine Learning, Computer Vision, TensorFlow, NumPy, Jupyter Notebook.

EDUCATION

Masters in Computer Science

2024-2026

Indiana University – Indianapolis

Bachelors in Computer Science and Engineering

2019-2023

Neil Gogte Institute of Technology, Hyderabad, India